SOLUTION REQUIREMENTS

Date	30 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Secure user login using username/password and session management.	High
2	Authorization Levels	Role-based access for **Admin, Teacher, and Student**.	High
3	External Interfaces	Integrates with **OpenCV, TensorFlow/Keras, MediaPipe, NLP/Speech APIs**, and report generation libraries.	High
4	Emotion Detection	Capture or upload face image, voice, or text and detect the user's emotional state using AI models.	High
5	Learning Recommendation	Provide personalized learning resources and suggestions based on detected emotions.	High
6	Reporting	Generate downloadable reports containing detected emotions, confidence scores, recommendations, and learning progress.	High
7	Business Rules	Accept only supported image, audio, and text formats. Emotion predictions are generated automatically.	Medium

S.No	Requirement Category	Requirement Description	Priority
8	Compliance	Ensure user privacy by securely storing emotional data and complying with data protection standards.	High
9	Dashboard & History	Display emotion history, learning progress, analytics, and previous reports.	Medium

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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Fast emotion detection and recommendation generation.	Response within **3–5 seconds** (excluding model latency).
2	Scalability	Support multiple concurrent users.	**100+ concurrent users**.
3	Security & Data Privacy	Protect uploaded images, audio, text, and emotion reports.	Encrypted storage and secure communication (HTTPS).
4	Reliability & Availability	System should remain available with minimal downtime.	**99% uptime**.
5	Accuracy	Emotion detection should provide reliable predictions.	**≥90% prediction accuracy** on the validation dataset.
6	Usability & Accessibility	Responsive and user-friendly interface compatible with multiple devices.	Works on **desktop, tablet, and mobile**.
7	Maintainability	Code should be modular, reusable, and easy to update.	Modular Python architecture with proper documentation.
8	Portability	Application should run across different operating systems.	Compatible with **Windows, Linux, and macOS**.